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Submitter Information

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General Comment

Dear Planning Committee,

Comments and recommendations from Rutgers University can be found in the attached document. We greatly appreciate the opportunity to provide feedback and would be glad to engage in further discussion if needed.

Thank you,
Michele Norin
Sr. Vice President & Chief Information Officer
Rutgers, The State University of New Jersey

Attachments

Comments on AI Action Plan from Rutgers University



Response to Request for Information: Development of an Artificial Intelligence (AI) Action Plan

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Submitted by Rutgers, The State University of New Jersey

March 7, 2025

Overview

Rutgers, The State University of New Jersey, recommends the following priorities for the development of an Artificial Intelligence Action Plan:

- Advance core AI research to ensure America is at the forefront of AI innovation
- Maximize AI's impact by applying AI to national priority areas, such as healthcare
- Enhance education and workforce development in AI to ensure American competitiveness and dominance
- Foster alliances between industry, government, and academia for national AI leadership
- Develop new infrastructure for data, computing, and manufacturing to secure America's position in AI innovation

Advancing Core AI Innovation to Solidify America's AI Leadership

While the current generation of AI technologies has demonstrated many remarkable achievements, there are many notable hurdles on the horizon for continuing the development of AI technologies that must be addressed through a coordinated national policy:

- *Improving Energy/Power Efficiency:* Implementing large-scale AI models, such as Large Language Models (LLMs), consumes large amounts of energy that will eventually burden U.S. power grid infrastructure.
- *Advancing AI Primitives:* Artificial General Intelligence (AGI) and Generative AI algorithms that use transformer-based methods have drawbacks in efficiency, scalability, generalizability, and adaptability that need to be addressed by researching alternative algorithm designs.
- *Broadening Data and Knowledge Encoding:* AI models are intrinsically dependent on their input data and internal representations of knowledge. Current AI models do not leverage the full variety of data sources nor efficiently represent knowledge gained from such data.
- *Developing Smarter AI:* Current AI algorithm approaches are adept at pattern matching and basic logic inferences, but they cannot perform more complex logic, such as abductive reasoning and hypothesis generation, when presented with partial data or complex-structured data.
- *AI that Continuously Learns:* Large-scale AI models involve frequent full-scale retraining on large input data, which can lead to outdated algorithms and potentially irrelevant or incorrect results.
- *Trustworthy and Reliable AI:* Many AI algorithms can be manipulated by adversarial information, and thus AI must be developed for mission-critical environments so that AI-based solutions are robust and resilient to errors, biases, and intentional hacking.

To address the above needs for advancing AI innovation in the U.S., Rutgers recommends the following actions as part of the nation's AI policy:

- *Develop National Technology/Innovation Competitions:* The nation should host competitions with specific technology objectives (e.g., minimizing computation usage on

benchmarks), with prizes promoting continued development and fostering commercialization.

- *Expand SBIR/STTR Program Scope:* Government funding agencies should evolve the Small Business Innovation Research (SBIR) and Small Business Technology Transfer (STTR) program model for AI innovation by encouraging public-private partnerships, providing more funding stages, streamlining the funding process, and removing hurdles between prototype and market deployment.
- *Create National Prototype Foundries:* The nation should create national chip foundries that facilitate onshore development of new AI hardware, such as chips that perform specialized AI computations for targeted applications.
- *Expand National Funding for Research Partnerships:* The U.S. should expand federal allocations to drive AI innovation by positioning universities to collaborate with applied research partners that support economic growth.
- *Encourage Innovation Through Collaboration by Establishing National Centers:* The U.S. needs to emphasize larger, team-based innovation by establishing national centers that bring together the best from government, academia, and industry to spur paradigm-shifting research.
- *Enhance Venture Capital Investments in AI Entrepreneurship:* The U.S. leads when it creates companies that alter the status quo of the technology landscape, and thus the private sector must be encouraged to invest in more U.S.-based AI startups.

Maximizing Impact by Applying AI to National Priorities

Artificial intelligence delivers the greatest value to the U.S. when solving critical societal challenges. Our university recommends a federal policy framework that prioritizes AI applications in high-impact areas where AI can enhance national security, economic competitiveness, and public well-being. Healthcare is a prime example, where AI can improve patient outcomes, reduce costs, and accelerate medical innovation, establishing the U.S. as a leader in this field.

The following policy actions apply across multiple sectors and are detailed elsewhere in this document. This section highlights how these actions can unlock AI's full potential in healthcare as an example of impact—showcasing how targeted federal investment and policy support can accelerate AI-driven breakthroughs, enhance patient outcomes, and position the U.S. as the global leader in AI-powered medicine.

Establish Dedicated Federal Funding for Applied AI Research

- *Advanced Methods:* Invest in next-generation AI methodologies, such as multimodal transformers and federated learning, to enhance AI's predictive capabilities and decision support capacity, which will revolutionize medical practices through earlier intervention.
- *Explainability:* Establish dedicated federal funding to develop advanced methods for transparent, interpretable AI that increases clinical trust and expands real-world deployment. This requires supporting interdisciplinary research integrating the most advanced techniques with expert domain knowledge.

- *Validation:* Fund pilot programs to evaluate AI-driven medical solutions under real-world conditions, allowing iterative improvements based on feedback.
- *Computing Infrastructure:* Ensure federally funded access to high-performance computing (HPC) resources, including GPUs and specialized AI accelerators, to support the development, training, and validation of complex AI models in healthcare. One promising option might be expanding the NSF ACCESS program and having other agencies replicate the program.

Example: AI-powered precision medicine requires large, broadly representative datasets and high-performance computing resources to train complex models for personalized diagnosis and treatment recommendations.

Expand Data Access & Interoperability for AI-Driven Applications

- *Data Access:* Expand the focus on creating data programs and centers to improve the accuracy and reliability of AI training datasets.
- *Interoperability and Standardization:* Support federal initiatives for health data interoperability (Fast Healthcare Interoperability Resources, Observational Medical Outcomes Partnership) to ensure AI models can integrate with electronic health records (EHRs), genomic databases, and clinical decision-support tools.

Example: AI-powered predictive healthcare models require access to secure, standardized data from multiple sources to generate accurate, real-time risk assessments for complex, chronic conditions like cardiovascular disease. Access to robust genomic, clinical, and real-world data ensures AI can accurately predict and optimize across populations.

Establish Adaptive Regulatory Pathways to Accelerate AI Deployment

- *Streamlined AI Approval Processes:* Develop adaptive, risk-tiered FDA pathways for AI-driven biomedical applications, diagnostics, and therapeutics, ensuring that regulatory processes match the pace of AI innovation.
- *AI Testbeds and Regulatory Sandboxes:* Establish federal AI testing environments where healthcare AI solutions can be piloted, refined, and validated under controlled conditions before broad adoption.

Example: AI-driven robotic-assisted surgery requires continuous regulatory adaptation as new AI models evolve to improve surgical precision, real-time decision support, and patient safety.

Expand Public-Private Partnerships for AI Initiatives

- *Research Collaboration:* Strengthen partnerships with universities and healthcare systems to accelerate AI development, validation, and clinical deployment.

- *AI Workforce Training and Education:* Establish federally supported AI training programs to equip clinicians, researchers, and regulatory professionals with the knowledge to leverage AI in medicine effectively.

Example: AI-driven automation in healthcare administration can reduce provider burnout by handling repetitive documentation tasks, allowing clinicians to focus on high-value patient care.

Prioritizing High-Impact Healthcare

Federal investment in AI should prioritize high-impact medical areas with strong potential for cost savings, improved patient outcomes, and broader technological applications:

- *Neurodegenerative Diseases (Alzheimer's, Parkinson's):* Support AI-driven early detection and disease modeling to slow their progression and reduce long-term care costs, which are projected to reach \$16.9 trillion globally by 2050.
- *Maternal-Fetal Medicine and Neonatal Care:* Support AI-powered risk prediction tools to reduce maternal and infant mortality rates and improve long-term population health outcomes.
- *Oncology:* Support AI-driven detection, treatment planning, and precision medicine to increase survival rates.
- *Cardiovascular and Metabolic Disorders:* Support AI prediction, enabling early intervention strategies that reduce healthcare costs.
- *Infectious Disease and Public Health:* Support AI-powered public health initiatives for real-time disease tracking and targeted intervention strategies, strengthening national preparedness and response.
- *Mental Health Diagnosis and Treatment:* Support initiatives to use AI to identify mental health challenges revealed in different kinds of data (text, behavior) and to prepare professionals to identify and deliver effective therapeutic interventions.

Scaling AI Innovation Across Sectors

Federal investment in AI should extend beyond healthcare to maximize cross-sector economic and national security benefits:

- *National Security:* AI-powered threat detection, cybersecurity automation, and military logistics optimization will enhance defense readiness.
- *Infrastructure Resilience:* AI can predict failures in critical infrastructure, optimize energy grids, and improve disaster response planning.
- *Environmental Sustainability:* AI-driven climate modeling, emissions monitoring, and resource optimization will support federal sustainability initiatives.
- *Education and Workforce Development:* AI-powered adaptive learning and workforce training models can prepare the next generation for an AI-driven economy.

Call to Action: The Need for Federal AI Investment

These strategic investments in applied AI will establish the U.S. as a global leader in explainable, high-impact AI and set the standard for innovation across sectors. In healthcare, AI will accelerate medical breakthroughs, enhance economic competitiveness, and drive the future of AI-powered medicine. By prioritizing advanced, mindful AI research and deployment, federal policy can transform patient care, fuel industry growth, and propel sustainable economic advancements.

Enhancing AI Education and Workforce Development to Sustain U.S. Dominance

The existing workforce is too small to meet the demand for technically skilled workers to develop and deploy the new AI and machine learning (ML) solutions that will ensure American leadership in these frontier areas. These necessary advancements in AI—and towards AGI—require basic research, often with collaboration among core AI disciplines, including computer science, statistics, applied math, and engineering. Collaboration with other application domains is also needed, especially those dealing with human intelligence and behavior.

The lack of computer science education in the U.S., particularly in high schools and undergraduate programs, hinders AI talent development. Policies are needed to ensure partnerships across industries. Universities should be supported to develop a highly skilled and competitive American workforce that will lead in the innovation and expansion of data science solutions. Leveraging the extension of these partnerships to local communities will expand the use and benefit of new data science solutions to local municipalities throughout the country.

We recommend the following policy actions:

- Incentivize the integration of understanding and use of AI throughout the workforce pipeline from K-12 to community college education programs.
- Incentivize universities to integrate AI education and training into undergraduate educational programs, ranging from English and the arts to STEM to professional programs (e.g., business).
- Incentivize universities to provide AI training through continuing education and extension programming to working professionals and community members, supporting a competitive American workforce that will effectively integrate the use of AI.
- Incentivize scholarships and internships for top students to pursue AI R&D to work with industry to develop the next generations of AI technology.
- Establish educational programs across the country that emphasize innovation and entrepreneurship to ensure that students have skills aligned with economic needs and solving emerging problems.

Strengthening Academic/Industry Collaboration to Foster AI Leadership

Innovation in AI is accelerating rapidly across the entire ecosystem of academia, national labs, large/medium-size companies, and startups. Research, education, and workforce development to ensure American leadership in AI require collaborative partnerships across institutions. Industry can partner with research universities for both teaching and research, providing access to data, resources, and the capability to seamlessly transition research to practice, while research universities can pursue cutting-edge lines of research that provide the intellectual innovation and foundation needed to advance the field. By working together, academia and industry can bridge the gap between theory and real-world application, ensuring that AI innovations push the boundaries of research and quickly translate into practical benefits for society.

We recommend the following policy actions:

- Develop a general framework for how industry professionals can collaborate with universities, STEM, and K-12 educators to help prepare the workforce for AI integration. Encourage academia-industry consortia to identify and understand industry priorities and needs for research and education.
- Incentivize integrated training partnerships between academia and industry, to enhance and revise curriculum and research (for example, following the NSF Industry-University Cooperative Research Centers, NSF Engineering Research Centers, NSF Science and Technology Centers, and NSF Research Traineeship models). This is particularly important in disciplines which will be transformed by AI but do not historically have a high degree of industry-aligned training, such as computational biology.
- Invest in collaborative AI training and innovation within academic research training environments, which can integrate a breadth of expertise and techniques. For example, investing in collaborative research requires both academic and industry participation. This will ensure that there is always a new generation of scientists, computer scientists, and engineers who are deeply motivated to capitalize on innovation opportunities.

Investing in Data and Computing Infrastructure to Secure America's AI Preeminence

Technology infrastructure is essential for advancements in AI. We recommend the following policy actions to ensure the U.S. retains its AI leadership:

- *Establish national AI research centers:* Funding large, centrally shared AI ecosystems will allow the federal government to maximize efficiency and reduce costs over locally run resources. Policy actions such as streamlining permitting, providing tax incentives, and incentivizing upgrades to power infrastructure will help the U.S. move quickly enough to maintain its leadership in AI research.

- *Maximize the value of existing research investments through data sharing:* Data is a fundamental raw material in recent AI development. With the generous support of the federal government, the U.S. conducts most of the research spending worldwide. All federal grants should cover the costs necessary to make this data available to U.S. AI researchers.
- *Invest in cloud computing:* The federal granting agencies should engage with U.S. public cloud providers to secure favorable pricing for access to compute, storage, and GPUs needed for AI. Leveraging federal purchasing power for these contracts will reduce the burden on the American taxpayer and make it easier for startup businesses to innovate and succeed.
- *Support AI hardware manufacturing:* Incentivize the domestic production of specialized AI hardware to ensure its availability to U.S. industry and academic institution.